

E-Commerce Sales & Customer Analytics

Industry-style data analytics project: complete guide (what, why, when, how, where) + interview Q&A

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Stack: Python, pandas, SQL (SQLite), matplotlib, reportlab

1. Project Overview

1.1 Business problem

An online retailer in India (Electronics, Fashion, Home, Beauty, Grocery, Sports) sells across 8 cities. Management asks: where does revenue come from, who are the most valuable and at-risk customers, do customers return, why are returns high, and what should we expect next quarter?

1.2 Objectives

- Clean and model raw transactional data into an analysis-ready warehouse
- Report core KPIs: revenue, orders, AOV, growth, returns, cancellations
- Segment customers with RFM and measure retention with cohorts
- Forecast revenue and give actionable recommendations

1.3 Architecture

- Raw CSVs -> `etl.py` (clean, validate, log) -> SQLite star schema (`fact_sales`, `dim_customer`, `dim_product`) -> SQL queries -> pandas analysis -> charts/tables -> auto-generated insights -> PDF
- Design rule: one script per stage, shared helpers in `utils.py`, SQL kept in `sql/queries.sql`, numbers never hand-typed (written to `metrics.json` and pulled into reports)

1.4 Key definitions (say these in the interview)

- **Net revenue** = quantity x unit price x (1 - discount%), counted only for Delivered orders
- **AOV** = net revenue / delivered orders; **Return rate** = returned orders / all orders
- **Active customer** = at least one delivered order

1.5 Stakeholders and use

- Management (growth), Marketing (win-back and loyalty), Operations (returns), Finance (forecast)

1.6 Two-minute pitch

I built an end-to-end analytics pipeline for an e-commerce business. I simulated realistic messy data, cleaned it with logged data-quality rules, modelled it as a star schema in SQL, and calculated KPIs. I segmented customers with RFM, found that a minority of customers drives about half the revenue, measured cohort retention, and built a backtested forecast. Everything runs with one command and produces a report with recommendations such as win-back campaigns, return reduction and category diversification.

1.7 Limitations (be honest)

- Data is simulated, so findings demonstrate method, not real market truth
- Forecast uses only 24 months; growth percentages are inflated by the early ramp-up
- No causal claims (observational data); recommendations should be A/B tested

2. Data and Schema

2.1 Raw files (data/raw)

- **customers.csv**: customer_id, signup_date, city, gender, age
- **products.csv**: product_id, product_name, category, unit_price (60 products, 6 categories)
- **orders.csv**: order_id, customer_id, order_date, payment_method (UPI, Card, COD, NetBanking, Wallet), status (Delivered, Cancelled, Returned)
- **order_items.csv**: order_id, product_id, quantity, discount_pct

2.2 Injected data problems (to show real cleaning skill)

- Duplicate orders, null payment methods, negative quantities, impossible ages (150), messy city text ("pune ")

2.3 Warehouse (data/processed/warehouse.db)

- Star schema: **fact_sales** (grain = one order line) joined to **dim_customer** and **dim_product**
- Grain matters: one row per order item, so count orders with `COUNT(DISTINCT order_id)` to avoid double counting
- Indexes on customer_id, order_date, category for fast filtering and grouping

2.4 Cleaning rules (etl.py)

- Drop duplicate order_id; fill null payment with "Unknown" (keep the row, the order is real)
- Remove quantity <= 0; fix ages outside 16-90 with the median; standardise city text (strip, title case); fill gender with "Unknown"
- Every rule counts affected rows into reports/data_quality.json (auditability)

3. Methodology: When, Why, How, Where

3.1 ETL and data cleaning

- **When:** always before analysis; every real dataset has duplicates, nulls, bad types.
- **Why:** wrong inputs give confident wrong answers (duplicates inflate revenue, outliers skew averages).
- **How:** pandas `drop_duplicates`, `fillna`, `to_datetime(errors="coerce")`, range filters, then load to SQL; log counts.
- **Where:** every analytics/BI/ML job; in production this runs on Airflow/dbt/ADF on a schedule.

3.2 Star schema and SQL

- **When:** repeated reporting on transactions; many questions on the same facts.
- **Why:** facts + dimensions make joins simple, queries fast, and metrics consistent.
- **How:** JOIN, GROUP BY, CASE WHEN, COUNT(DISTINCT), window RANK() OVER (PARTITION BY ...).
- **Where:** data warehouses (Snowflake, BigQuery, Redshift, Postgres) feeding Power BI/Tableau.

3.3 KPI analysis

- **When:** monthly business reviews, dashboards. **Why:** KPIs turn raw rows into decisions (growth, margin proxies, risk).
- **How:** monthly aggregation, AOV, MoM growth with `pct_change`, category/city splits, return and cancel rates.
- **Where:** exec dashboards, finance and ops reviews.

3.4 RFM segmentation

- **When:** you need to prioritise marketing spend among customers. **Why:** customers are not equal; recency, frequency, monetary predict value and churn risk cheaply and explainably.
- **How:** Recency (days since last order, quintiles, recent = high), Frequency (business bins clipped 1-5 because many ties break quantiles), Monetary (quintiles); rules map scores to Champions, Loyal, New, At Risk, Lost.
- **Where:** CRM, email/push campaigns, loyalty programs, retail and banking.

3.5 Cohort retention

- **When:** to learn whether customers come back and whether newer cohorts are better. **Why:** averages hide churn; cohorts separate acquisition month from behaviour over time.
- **How:** cohort = first purchase month; index = months since; retention = active customers / cohort size.
- **Where:** SaaS, e-commerce, apps, subscription businesses; product and growth teams.

3.6 Forecasting

- **When:** planning stock, ad budget, targets. **Why:** decisions need numbers about the future with a known error.
- **How:** raw trend models failed here (ramp-up growth gave a negative forecast), so revenue was normalised per registered customer, last year's seasonal shape was applied to the latest level, and error was measured by backtesting (MAPE).
- **Where:** demand planning, finance, capacity planning. Upgrade path: Holt-Winters, SARIMA, Prophet, gradient boosting with more history.

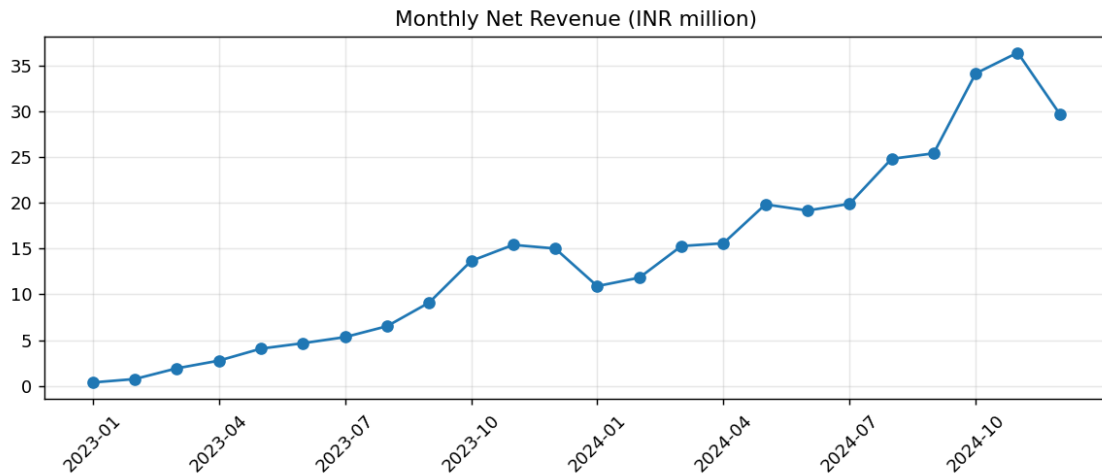
3.7 Tool choices

- **pandas** for in-memory analysis up to a few million rows; move to Spark/SQL engines beyond that.
- **SQLite** for a zero-setup demo; in production use PostgreSQL/Snowflake (same SQL).
- **matplotlib** for static charts; use Power BI/Tableau for interactive dashboards.
- **Auto-generated docs** guarantee report numbers match the data.

4. Results, Insights and Recommendations

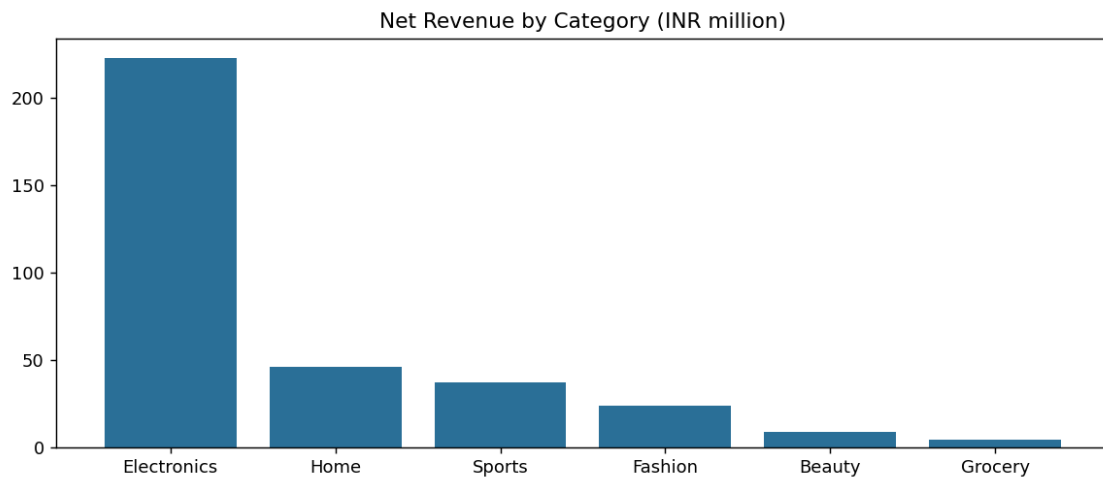
4.1 Headline KPIs (delivered orders only)

- Net revenue: **INR 342.2M** from **18,202** delivered orders
- Average order value (AOV): **INR 18,798**; active buying customers: **4,370**
- Return rate **6.2%**, cancellation rate **4.0%**
- Peak month: **2024-11** (INR 36.4M) - festive-season effect
- Year-on-year growth: **230.5%** (CAUTION: inflated because the customer base was still ramping up in 2023; a mature business would be far lower)



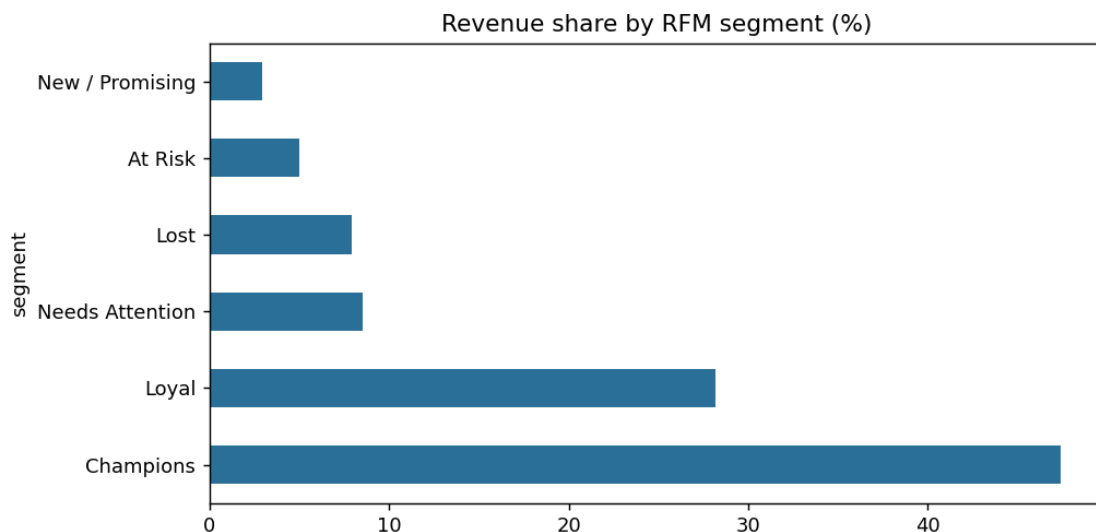
4.2 Where the money comes from

- Top category: **Electronics** = 65.1% of revenue (concentration risk)
- Top city: **Mumbai** = 20.9% of revenue
- Highest return-rate category: **Sports**
- **Electronics**: revenue INR 222.9M, return rate 6.0%, avg discount 6.6%
- **Home**: revenue INR 45.9M, return rate 5.8%, avg discount 6.6%
- **Sports**: revenue INR 36.9M, return rate 6.6%, avg discount 6.6%
- **Fashion**: revenue INR 23.5M, return rate 6.3%, avg discount 6.4%
- **Beauty**: revenue INR 8.9M, return rate 6.0%, avg discount 6.6%
- **Grocery**: revenue INR 4.1M, return rate 6.2%, avg discount 6.4%



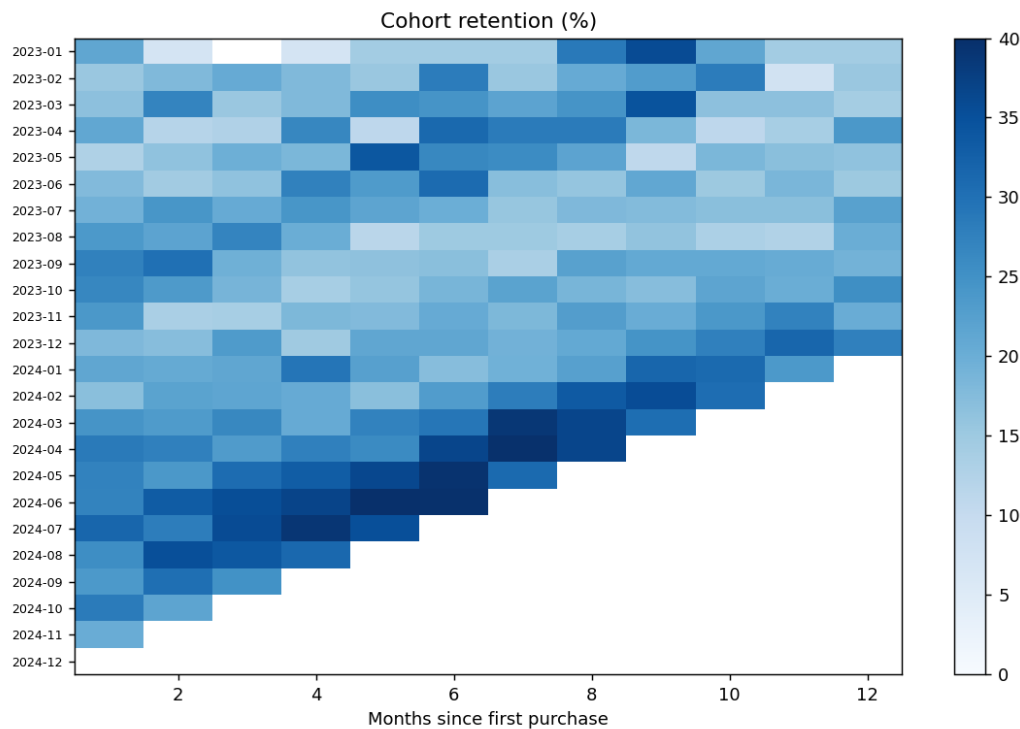
4.3 Customer segments (RFM)

- **Champions:** 1189 customers (27.2%), 47.4% of revenue, avg value INR 136,372
- **Loyal:** 896 customers (20.5%), 28.2% of revenue, avg value INR 107,843
- **Needs Attention:** 647 customers (14.8%), 8.5% of revenue, avg value INR 45,107
- **Lost:** 1018 customers (23.3%), 7.9% of revenue, avg value INR 26,613
- **At Risk:** 304 customers (7.0%), 5.0% of revenue, avg value INR 56,477
- **New / Promising:** 316 customers (7.2%), 2.9% of revenue, avg value INR 31,472
- Champions are 27.2% of customers but **47.4%** of revenue (Pareto effect)
- At Risk customers (304) hold 5.0% of revenue: cheapest win-back target



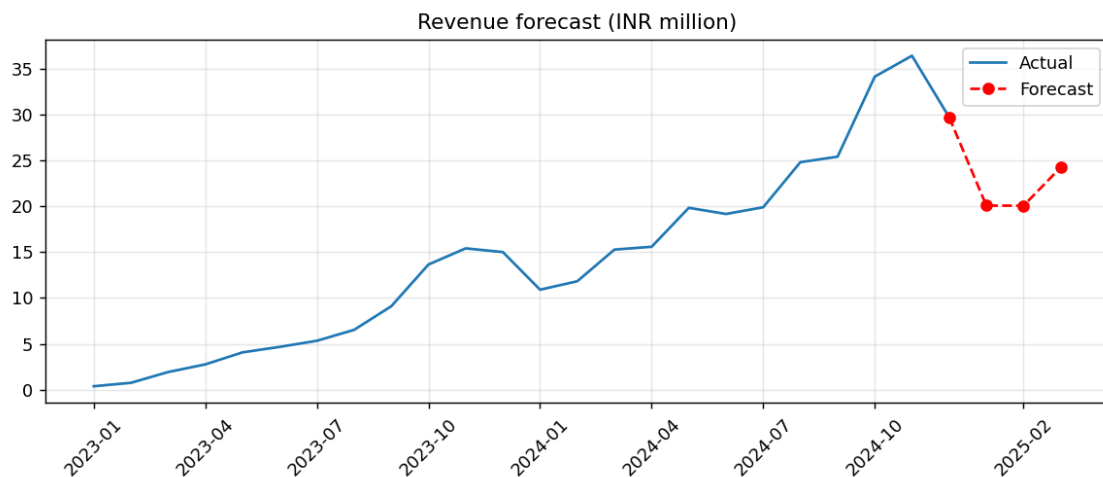
4.4 Retention (cohorts)

- Average month-1 retention: **22.6%**; month-3: **23.1%**
- Reading: customers who do not come back in month 1 rarely return later, so onboarding and the first 30 days matter most



4.5 Forecast (next 3 months, starting 2025-01)

- 2025-01: INR 20.1M
- 2025-02: INR 20.0M
- 2025-03: INR 24.2M
- Backtest error (MAPE on last 3 known months): **3.4%**. Seasonal dip after the festive peak is expected.



4.6 Recommendations

- **Diversify:** Electronics drives 65.1% of revenue; grow Home/Beauty/Fashion to reduce single-category risk.
- **Reduce returns:** investigate Sports (size/quality/description issues); each 1 point of return rate recovers revenue and logistics cost.
- **Win-back campaign:** target the At Risk segment with personalised offers before they become Lost.
- **Protect Champions:** early access, loyalty tiers; they carry nearly half of revenue.
- **Fix month-1 retention:** welcome journey, second-purchase coupon within 30 days.

- **Plan for seasonality:** stock and ad budget ahead of Oct-Dec; expect a Q1 dip.

4.7 Data quality log (ETL)

- Raw orders 20,469; duplicates removed 203; null payment methods filled 409
- Invalid quantities removed 151 of 50,460 item rows; age outliers fixed 50; final fact rows 50,309

4.8 Resume bullets

- Built an end-to-end e-commerce analytics pipeline (Python, SQL, SQLite star schema) on 18,202 orders and INR 342.2M revenue, with automated cleaning and data-quality logging.
- Segmented 4,370 customers with RFM, showing 27.2% of customers drive 47.4% of revenue, and quantified cohort retention and return hotspots.
- Delivered a customer-base-normalised revenue forecast with 3.4% backtest MAPE plus an auto-generated PDF report and business recommendations.

5. Interview Questions and Answers

5.1 Project and business

Q1. Explain your project in 1 minute.

End-to-end e-commerce analytics: simulated messy transactional data, cleaned and loaded into a SQL star schema, KPIs, RFM segmentation, cohort retention, a backtested forecast, and a report with recommendations. One command reproduces everything.

Q2. Why e-commerce and why this data?

It has all classic analytics problems: transactions, customers, products, returns, seasonality. Simulation let me inject realistic dirt and know the truth, so I could validate my methods.

Q3. What business value did it produce?

It shows revenue concentration (category and customer), a high-value Champions segment, at-risk customers to win back, return hotspots, weak early retention, and a forecast with known error, each tied to an action.

Q4. How would you present to a non-technical client?

Lead with 3 headline numbers and 3 actions, use the charts, avoid jargon, explain RFM as "best, at-risk and lost customers", and state limits.

Q5. What are the project limitations?

Simulated data, 24 months of history, observational (no causality), growth inflated by ramp-up, simple forecast. With real data I would add margin, marketing spend and A/B tests.

5.2 Data cleaning and ETL

Q6. What data quality issues did you handle?

Duplicate orders, null payment methods, negative quantities, impossible ages, inconsistent city text. Each rule logs how many rows it touched.

Q7. Why fill payment nulls instead of dropping rows?

The order is real revenue; dropping it would understate sales. Labelling "Unknown" keeps the row and the gap visible.

Q8. Mean or median to fix outliers?

Median, because it is robust to outliers; the mean is pulled by the very values being fixed.

Q9. How do you avoid double counting?

Fact grain is one order line, so orders are counted with `COUNT(DISTINCT order_id)`, and revenue is summed at line level.

Q10. Why exclude Cancelled and Returned from revenue?

Net revenue should reflect money actually kept. Returns are still measured separately as a risk metric.

Q11. How would you productionise the ETL?

Schedule with Airflow or dbt, add tests (uniqueness, not-null, ranges), incremental loads, alerting, and version control.

5.3 SQL

Q12. WHERE vs HAVING?

WHERE filters rows before grouping; HAVING filters groups after aggregation.

Q13. INNER vs LEFT JOIN?

INNER keeps matches only; LEFT keeps all left rows with NULLs where no match (useful to find customers with no orders).

Q14. Where did you use a window function?

RANK() OVER (PARTITION BY city ORDER BY revenue DESC) to find the top category per city without a self-join.

Q15. COUNT(*) vs COUNT(DISTINCT x)?

COUNT(*) counts rows; COUNT(DISTINCT x) counts unique values, essential at line-level grain.

Q16. How do you speed up a slow query?

Index filter/join columns, select only needed columns, filter early, avoid functions on indexed columns, check the query plan, pre-aggregate.

Q17. Why a star schema?

Simple joins, consistent metrics, fast aggregates, and easy BI connection compared with a flat or highly normalised design.

5.4 Analysis methods

Q18. What is RFM and why use it?

Recency, Frequency, Monetary scoring of customers. It is simple, explainable and effective for prioritising marketing.

Q19. Why did you not use quantiles for Frequency?

Most customers have 1-3 orders, so quantiles create ties and arbitrary splits. I used business bins clipped at 5.

Q20. How do you pick RFM segment rules?

From business meaning (e.g. high R and high F = Champions), then check sizes and revenue share, and adjust. K-means is an alternative but less explainable.

Q21. What is a cohort analysis?

Group customers by first purchase month and track the share still buying in later months. It reveals retention beyond blended averages.

Q22. What did retention tell you?

Month-1 retention is low and flat afterwards, so the first 30 days decide loyalty; the action is onboarding and a second-purchase incentive.

Q23. How did you forecast and how did you validate?

Linear and log trends failed because of ramp-up growth. I normalised revenue per registered customer, applied last year's seasonal shape to the latest level, and backtested on the last 3 months with MAPE.

Q24. What is MAPE and its weakness?

Mean absolute percentage error. Easy to interpret but unstable near zero values and treats over/under forecasts equally.

Q25. Why is YoY growth very high?

The business was still acquiring customers in 2023, so the base year is tiny. I flagged it rather than presenting it as organic growth.

Q26. Correlation vs causation here?

Data is observational: segments and retention show association, not proof. I would run A/B tests on win-back offers.

5.5 Scenario and judgement

Q27. Revenue dropped 15% last month. What do you do?

Check data first (missing loads, definitions), then decompose by orders x AOV, category, city, channel, new vs returning, and compare with last year's seasonality; then form hypotheses.

Q28. Marketing has a small budget. Whom do you target?

At Risk customers with high past value (cheapest win-back), protect Champions with loyalty perks, and avoid spending on Lost customers.

Q29. How would you scale this to 100 million rows?

Move to a warehouse (BigQuery/Snowflake) or Spark, push aggregation to SQL, partition by date, and schedule incremental pipelines.

Q30. What would you improve next?

Margin and CAC data, customer lifetime value, churn prediction model, proper time-series models, an interactive dashboard (Power BI/Streamlit), and CI tests for the pipeline.

Q31. How do you ensure results are reproducible?

Fixed random seed, one-command pipeline, SQL in version control, metrics stored in JSON and inserted automatically into the report.

Q32. Which chart type for which message?

Line for trends, bar for category comparison, heatmap for cohorts, horizontal bar for ranked shares. Always title, units and one takeaway.